

CSE-481 Engineering Economics

Week 2

**Agentic Harness Output:
Result + Why? + Risks + Human Review**

BIST 100 Financial Analytics Project

Illustrative Crisis Pattern (Generic)

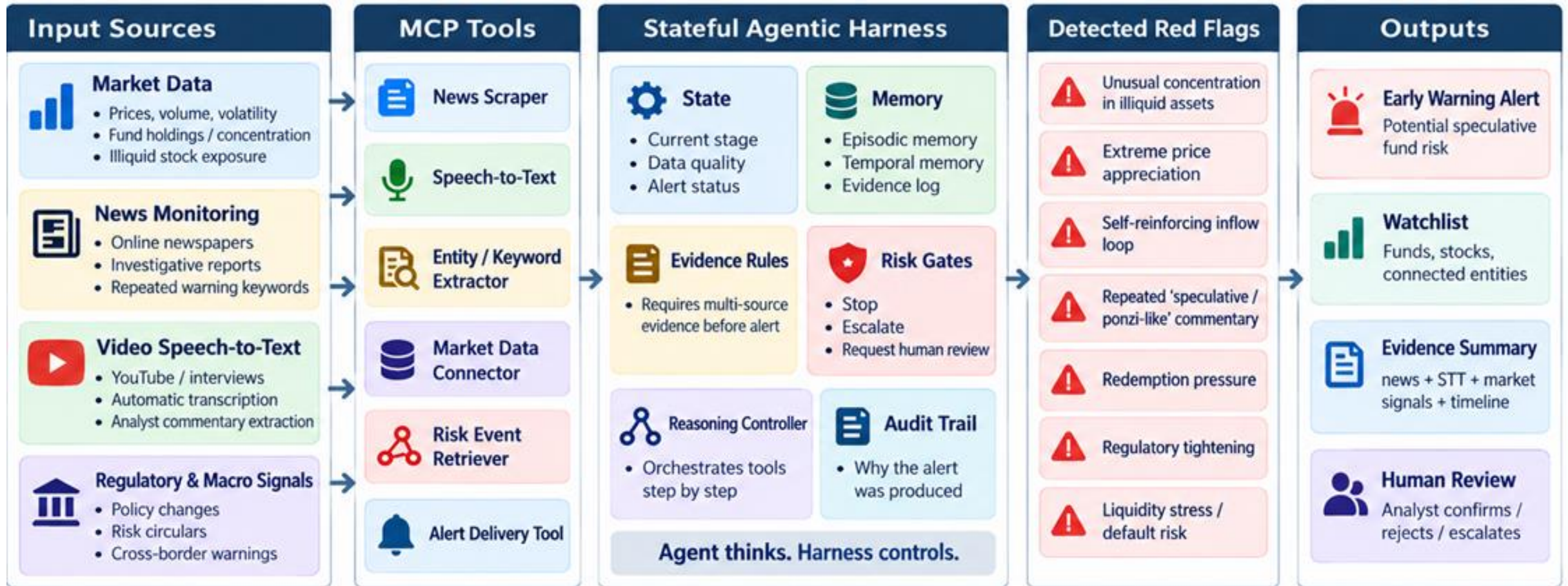


Why the Harness Matters

- ✓ Listens to multiple information channels
- ✓ Captures repeated warnings earlier
- ✓ Combines qualitative and quantitative evidence
- ✓ Produces explainable alerts instead of unsupported guesses

Agentic Harness for Early Detection of Speculative Fund Risk

Integrating market data, news, speech-to-text, and regulatory signals



What is a Harness in Agentic AI?

**A harness is the control layer around an AI agent.
It manages steps, tool permissions, evidence, risk gates, memory, logging and human review.**

Agent thinks.

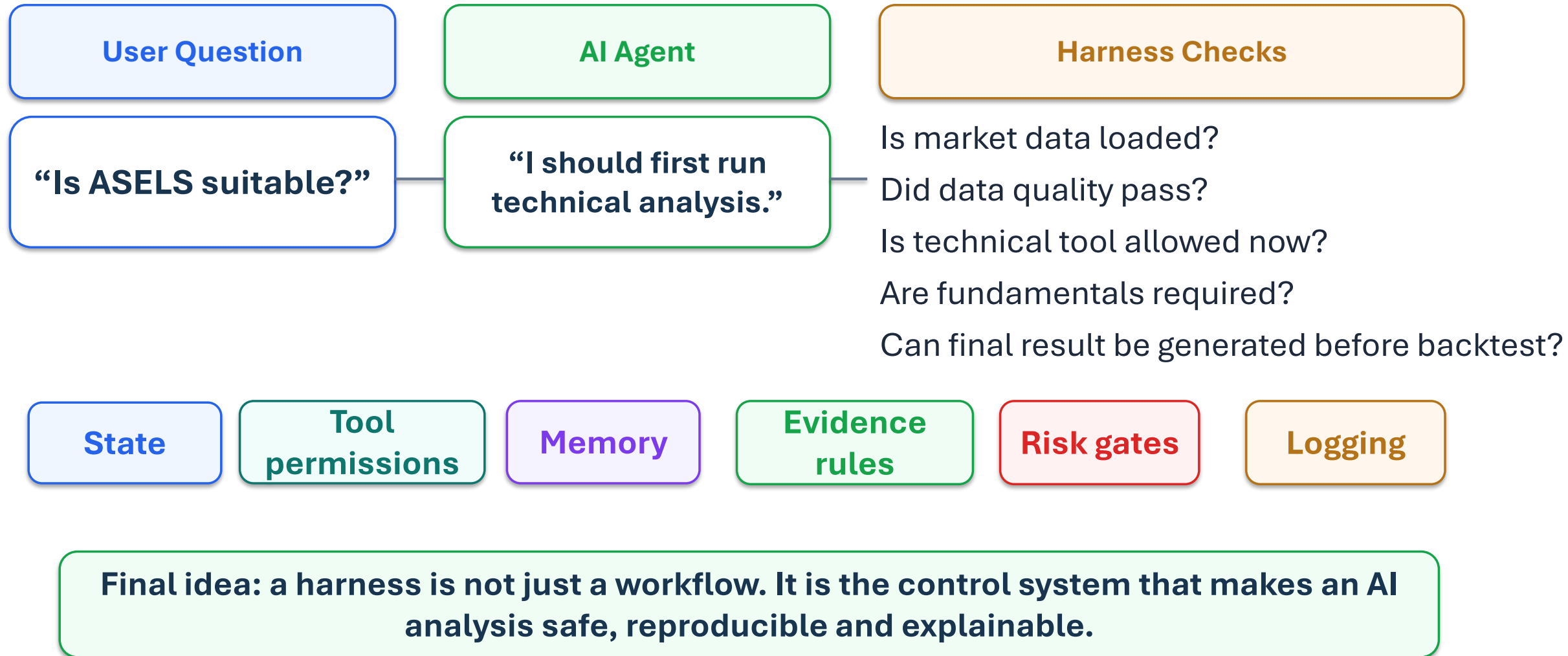
Harness controls.

Agent interprets:
“I should perform technical analysis first.”

Harness checks:
“Are we in the right state? Is this tool allowed?”

**Short definition: The agent decides what seems useful; the harness decides what is allowed next.
The role of the harness is to make that reasoning controlled, ordered, and safe.
The agent reasons about what to do. The harness controls what it is allowed to do.**

BIST Example: “Is ASELS Suitable?”



Integrated BIST 100 Agentic Financial Harness

From market data to evidence-based educational trading signals

BIST 100 stocks
Technical analysis + sector analysis
Fundamentals from Fintables
Macro data, news and speech context
MCP tools + stateful AI harness
Backtesting + evidence + human review

What is the goal?

**Not to guess prices with magic.
Build a controlled system that analyzes,
verifies and explains.**

Data

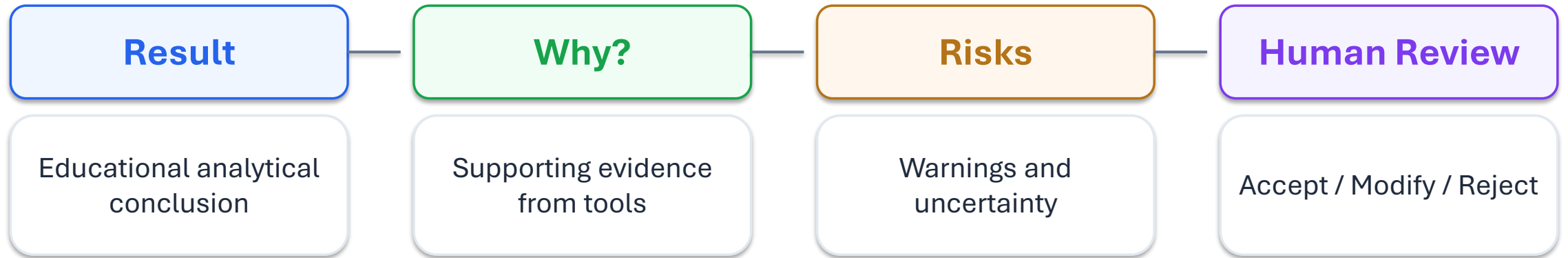
Tools

Harness

Backtest

**Final output:
Educational analytical conclusion
+ evidence + warnings + human review**

What Should the Final Output Include?



**The system should not output only “BUY” or “SELL”.
It should explain what it found, why it found it, what the risks are, and how the human reviewer responded.**

Educational analytical conclusion: not investment advice; it is a research/analysis label.

Evidence must come from deterministic Python tools, Fintables data, backtest results or verified context.

Warnings show uncertainty, weak samples, missing data, high volatility or historical limitations.

Example Final Output for a BIST Stock

Educational Analytical Conclusion

POTENTIAL_CATCH_UP_CANDIDATE
Example: THYAO

Evidence

Stock is 8% behind sector median.
KAMA slope turned positive.
Revenue Growth improved.
Backtest outperformed XU100.

Warnings

Only 12 similar historical cases.
Market volatility is high.
Historical performance is not a future guarantee.

Human Review
REJECT

Reason: Maximum drawdown is too high.

Result + Why? + Risks + Human Review

Key idea: the AI agent explains a reproducible analysis; it does not make the final decision.

Why Do We Need an Agentic Harness?

Without a harness

- The LLM may guess RSI, P/E (Price-to-Earnings Ratio) or ROE (Return on Equity).
- The workflow may skip validation.
- Different runs may give different conclusions.
- No clear evidence chain is preserved.
- Backtesting may be forgotten or done incorrectly.

Example mistake:

"THYAO RSI is probably around 60"

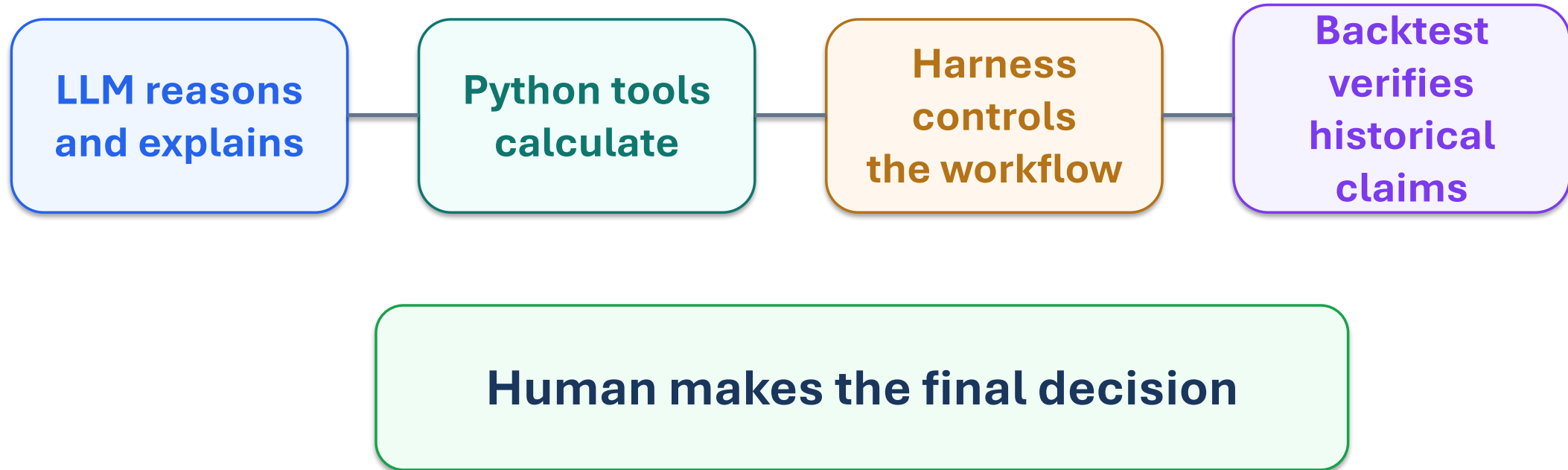
With a harness

- The LLM must call deterministic tools.
- The workflow follows fixed states.
- Evidence is stored and replayable.
- Unsafe analyses can be blocked.
- Backtesting verifies historical claims.

Correct pattern:

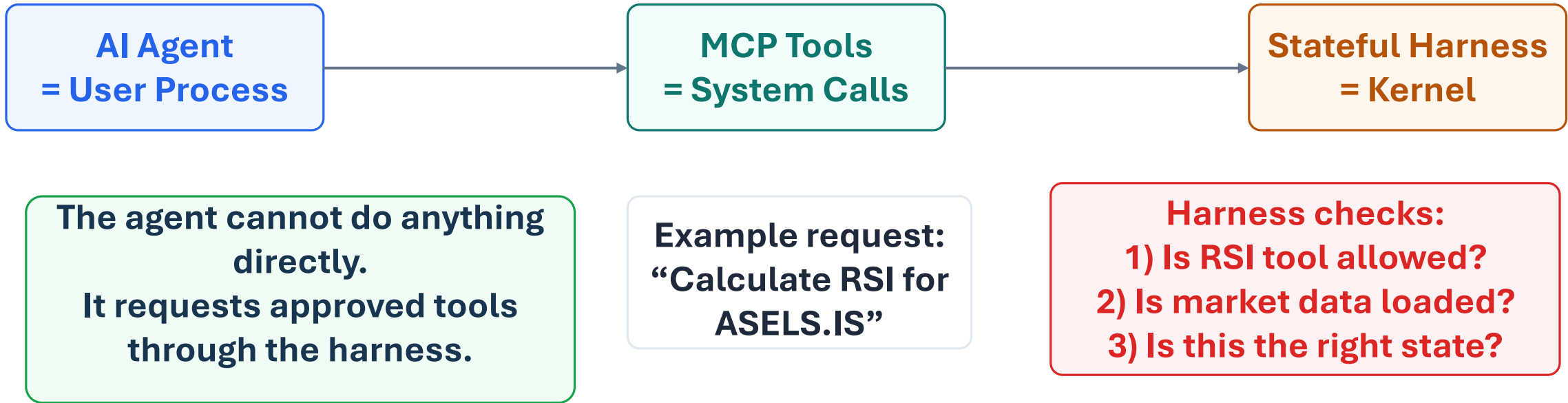
**LLM asks tool → tool calculates RSI →
harness logs result**

The Main Principle



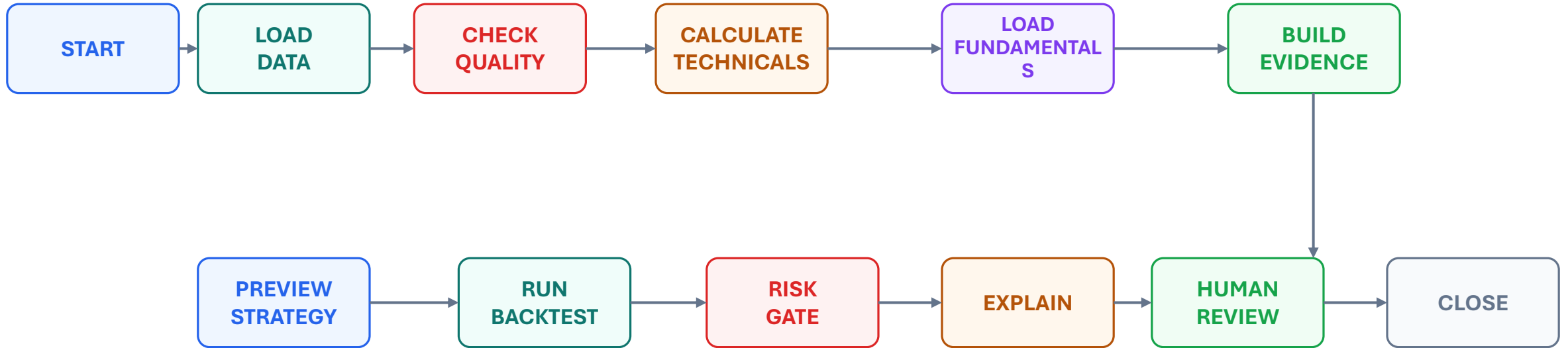
Example: The LLM should never estimate THYAO's RSI, ASELS's ROE, or TUPRS's Net Debt/EBITDA. The AI agent orchestrates analysis. The numerical facts must come from deterministic tools.

Metaphor 1 — Harness as an Operating System Kernel



OS analogy: processes use system calls; agents use MCP tools.
The kernel controls permissions; the harness controls allowed actions.
State prevents the agent from jumping to unsupported conclusions.

Metaphor 2 — Harness as a Finite State Machine



**Stateful means the system remembers:
current state + completed steps + available evidence + allowed tools**

BIST example: if Backtest = NOT RUN, the FSM does not allow a final strong conclusion.

Metaphor 3 — Harness as a CI/CD Pipeline for AI Reasoning



CI/CD idea:
A system should not
deploy if tests fail.

Financial harness idea:
The AI should not generate a strong
conclusion if data quality or
backtest verification fails.

Example failure:
Future financial-statement
data appears inside a
historical backtest → **HARD
BLOCK**

- Each stage produces an artifact (generated output): data table, indicator table, evidence bundle, backtest report, warning list.
- The final explanation is like a release artifact: it must be reproducible and traceable.
- AI reasoning also needs engineering controls.

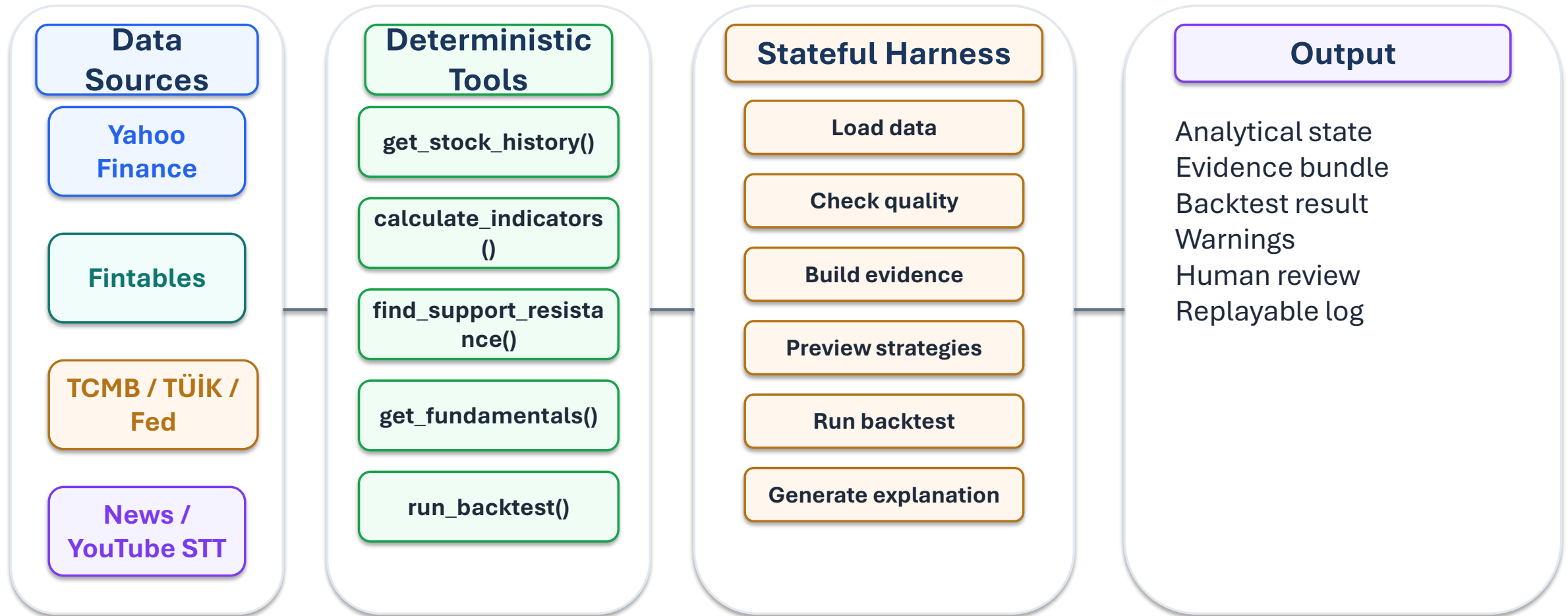
Connecting the Metaphors to an AI Agent for BIST Analysis



The agent can reason, but the harness decides what it is allowed to do next.

Stateful = the system remembers where it is, what it has done, and what must happen before the next step.

Big Picture: Data → Tools → Harness → Decision



Specialized Agents in the Project

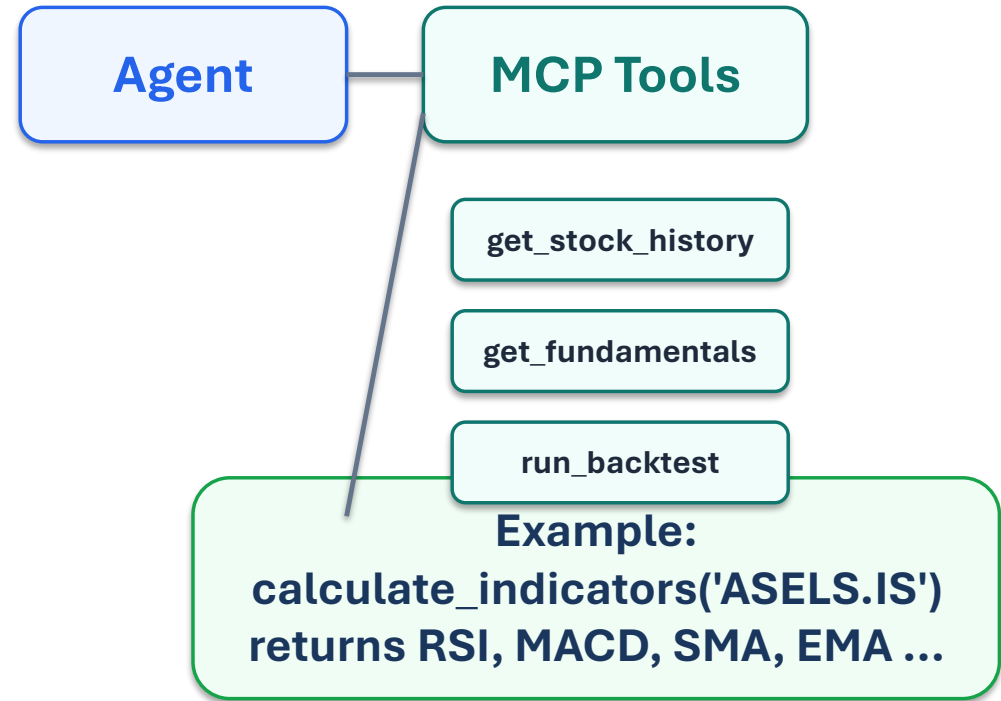


**Idea: not every data source becomes an agent.
Reasoning agents interpret. Deterministic services calculate.**

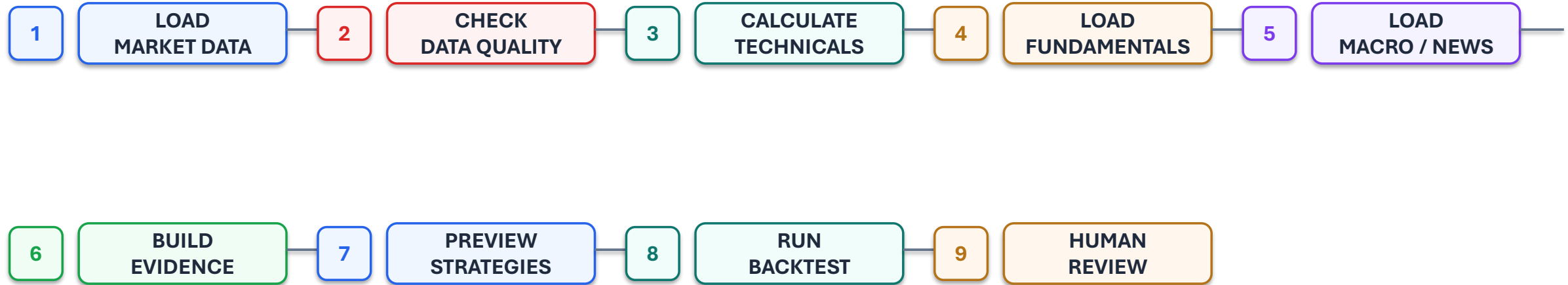
What is MCP in This Project?

MCP (Model Context Protocol)
lets the AI agent call approved tools in a
controlled way.

The agent does not run arbitrary code.
The agent sees only allowed tools.
Each tool has a known input and output schema.
This makes the workflow safer and more
reproducible.



Stateful Harness: The Step-by-Step Workflow



Important idea:
The agent cannot jump directly to a final conclusion before data validation and backtest verification.

Memory in a Financial Agent

Temporal Memory

What changed over time?
Prices, volume, RSI, ROE, FX,
inflation

BIST example:
ASELS RSI moved from
31 → 44 → 57.
That history belongs to
temporal memory.

Episodic Memory

What happened in past
analyses?
State → signal → backtest →
human decision

BIST example:
Last month the system
marked THYAO as
MONITOR/WATCH.
Human later rejected it
because drawdown
was high.

Procedural Memory

What is the correct workflow?
Data → validation → analysis →
backtest → explanation

BIST example:
The system knows it
must run the risk gate
before generating the
final explanation.

Temporal vs Episodic vs Procedural Memory

Temporal

Stores time-series facts.

Example:
TUPRS quarterly EBITDA
growth changed across
four quarters.

Episodic

Stores past cases.

Example:
“AKBNK lagged its sector,
then caught up after a
successful backtest.”

Procedural

Stores steps and rules.

Example:
“Never use future
financial statements
inside a historical
backtest.”

Simple rule:

Temporal = data history • Episodic = case history • Procedural = workflow history

Evidence Bundle and Committed Evidence

Observed Context

RSI, MACD, SMA, EMA
Sector lag score
Revenue growth
ROE and leverage
USD/TRY and inflation
News events
Speech topics

Committed Evidence Set

E1: Stock is 8% behind sector median.
E2: KAMA slope turned positive.
E3: Revenue growth improved.
E4: Lower Bollinger reversals worked in 11/17 similar cases.
E5: Backtest variant B outperformed XU100.

Idea: the system may see many features, but it should clearly state which ones actually support the final conclusion.

Scenario 1 — Sector Laggard Catch-Up

Question

**Inside the same sector,
which stock is lagging now
but may catch up later?**

Build sector peer groups.
Rank leaders and laggards.
Check if laggards later outperform.
Filter with fundamentals and technical state.

Illustrative banking example

GARAN

+16%

AKBNK

+12%

ISCTR

+11%

YKBNK

+3
%

Interpretation:

YKBNK is lagging the sector. The project tests whether such laggards historically catch up.

Scenario 2 — Weekday and Multi-Day Patterns

Question

**Does a stock show patterns
such as
Mon–Tue–Wed down and
Thu–Fri up?**

Mon
-1.2%

Tue
-0.6%

Wed
-0.4%

Thu
+0.8%

Fri
+1.0%

Count how often the pattern appears.
Measure average/median returns.
Test stability out of sample.
Include transaction costs.

Important warning:
**Some patterns appear by chance. The project
must test whether they survive later periods.**

Scenario 3 — Technical Reversal Events

Bollinger Reversal

Price touches lower band
then bounces upward

Example:
24 lower-band
touches
15 successful 5-day
rebounds

RSI Trough / Peak

RSI local low or high
before reversal

Example:
RSI local trough +
volume rise
may strengthen a
reversal

Supertrend / KAMA / Ichimoku

Trend flip or cloud breakout
with follow-through

Example:
Bullish Supertrend
flip
above the cloud

Measure how many times these events occurred and how strong the forward reversal was.